

Step-5 Overview

Step-5 Overview

This section provides a **brief overview of what step-5 accomplishes**. Like previous steps, the program is **dimension-independent** thanks to the `template<int dim>` class design. Its key feature is that it allows computation of the solution on **successively refined grids** automatically.

Key Features of Step-5

1. Reading a grid from disk

The program demonstrates how to load a pre-generated mesh from an external file instead of generating it programmatically.

2. Handling non-constant coefficients

The PDE now includes a spatially variable coefficient, for example:

$$a(x) = \begin{cases} 20 & \text{if } |x| < 0.5, \\ 1 & \text{otherwise.} \end{cases}$$

3. Preconditioning the Conjugate Gradient (CG) solver

Step-5 uses the **SSOR preconditioner** with a relaxation factor of 1.2. SSOR typically performs better with specific DoF enumerations (as seen in step-2), though enumeration is not explicitly applied in this step.

4. Support for debugging via assertions

The program uses `Assert` statements to ensure **safe programming**. These assertions verify that parameters and internal states are valid in debug mode. For example:

```
1 Assert(dim == 2, ExcNotImplemented());
```

This prevents the program from running in unsupported dimensions and helps catch errors early.

5. Solving on successively refined grids

Unlike step-4, the `make_grid()` function is **not part of the main class**. Instead, the program:

- Reads the initial grid from a file inside the `run()` function
- Refines the mesh in a loop

- Calls `setup_system()`, `assemble_system()`, `solve()`, and `output_results()` within the loop

This approach allows computation and output of the solution at each refinement level automatically.

Summary

Step-5 demonstrates how to extend the previous tutorials by:

- Working with **externally generated meshes**
- Handling **non-constant PDE coefficients**
- Applying **preconditioning** in solvers
- Using **assertions for robust debugging**
- Automating **Output on successive mesh refinements**